2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), 13-15 February 2025, CUET, Chattogram-4349, Bangladesh

Automated Attendance system for the Entire Classroom with Multi-Face Facial Recognition via AWS Rekognition

MD KAMRUZZAMAN
Department of Electrical and Electronic Engineering
Chittagong University of Engineering and Technology
Chittagong, Bangladesh
mdkamruzzamansiam2015@gmail.com

Md. Ebnul Azuz Sheikh
Department of Electrical and Electronic Engineering
Chittagong University of Engineering and Technology
Chittagong, Bangladesh
ebnulajuj550@gmail.com

Md. Radowan Hossain
Department of Electrical and Electronic Engineering
Chittagong University of Engineering and Technology
Chittagong, Bangladesh
radowanhossain155@gmail.com

Muhammad Masum Billah Mojumdar
Department of Electrical and Electronic Engineering
Chittagong University of Engineering and Technology
Chittagong, Bangladesh
mbillah1994@gmail.com

2025 International Conference on Electrical, Computer and Communication Engineering (ECCE) | 979-8-3503-5750-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ECCE64574.2025.11013462

***Abstract*— Taking attendance is crucial for any class or meeting to account for a person's presence on the spot. Manual attendance methods are time-consuming, disruptive, and prone to proxy cheating. Existing methods such as QR scanning, iris recognition, and fingerprint recognition take more time and create congestion during the scanning process. Modern society requires an advanced method for taking attendance that is easily implementable, accurate, faster, and non-disruptive to any program. This study presents an advanced cloud-based attendance system with face recognition using the AWS API, which can provide accurate results from a single image containing multiple faces. The system utilizes AWS Rekognition for face detection and matching, Amazon S3 for secure storage of user images, and AWS Lambda for executing logic in an environment. DynamoDB serves as a database for storing registered student IDs and face prints and is enhanced by a user-friendly front-end interface to facilitate real-time interactions. Our designed model has been tested using various real-time images and has achieved 100% accuracy. It provides the total number of detected faces, indicating how many are present and how many are absent in the image, along with the number of unknown faces and their images. The proposed system is easy to implement, faster, more cost-effective, a time saver, and hassle-free to operate.**

Keywords—Face recognition, Face detection, CNN, AWS, Attendance system.

I. INTRODUCTION

Taking attendance in classrooms of universities, colleges, schools, any meeting or seminar room is necessary because it is related to academic performance, ensures accountability, encourages discipline, improves safety and security, and so on [1]. In the traditional method, teachers or hosts take attendance manually by calling the ID's or names of each student one by one. It makes interruptions and takes a long time. It is subjected to human errors like missing or incorrect entries, and there is a possibility of cheating by proxy attendance. There are also some updated methods available, including QR code scanning and fingerprint recognition. Cheating is possible in QR ID card scanning by using someone else's ID card when the specified person is absent. It also takes a long time [4]. In the fingerprint approach, using a proxy can be prevented, but it consumes more time and creates significant traffic at the scanning point. Consequently, several institutions have started using various attendance-taking methods such as Radio Frequency Identification (RFID), iris recognition, biometric identification, and others, which help to save time. The RFID method can detect multiple people simultaneously, which reduces time consumption, but it needs physical tags (ID cards). Tags can be lost or damaged and are often assigned to the wrong person, which can lead to cheating [5]. Face detection is a biometric feature that addresses issues of time consumption, missed entries, and cheating. It is more precise and quicker than other techniques and can effectively operate with various facial expressions. In this method, predefined faces are registered in a database. Face recognition from a single image consisting of multiple faces is done in two steps. First, the detection of faces from the image and then matches the detected faces with predefined faces from the database [6][7]. Based on this, this paper presents an automated cloud-based attendance system using face detection that utilizes Amazon Wave Services (AWS) [8] [9]. The work is as follows:

- Creating a system to automate attendance tracking using a single picture containing multiple faces.
- Recording the details of registered individuals and automatically track attendance in a database.
- Sending an email to the teacher's email of the recorded attendance.

The system introduces an optimized approach to attendance tracking by using facial recognition technology, making the process seamless and requiring just a single click to complete. Our system uses the AWS Rekognition API [10]

979-8-3503-5750-9/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: Chittagon University of Engineering and Technology. Downloaded on July 15,2025 at 21:54:09 UTC from IEEE Xplore. Restrictions apply.

[11] to recognize each student's face and efficiently log their attendance. This approach stands out by automatically recognizing all students' faces in a single image capture, offering a smoother and more convenient experience for both teachers and students.

II. Literature Review

Various methods have been used for implementing an attendance system. An eigen faces technique was used by Kar et al. [12], utilizing a Haar cascade classifier with the Paula-Vila algorithm for extracting characteristics and performing the analysis of the principal components. It classifies the faces from the saved training data at which index are available of the faces. However, it is sensitive to the orientation of the face. Arsenovic et al. [13] developed a method by combining several algorithms together. They employed a deep convolutional neural network (DCNN) that utilizes a CNN cascade for feature detection, a FaceNet CNN for face embedding, and a support vector machine (SVM) to categorize the faces into a pre-registered index, obtaining an accuracy of 95%. They needed training datasets captured at various angles, including frontal, upward, downward, and sideways tilts. However, the challenges of managing illumination and angular constraints during data collection, along with the system's sensitivity to these factors, made this approach unsuitable for real-world applications. An RFID-based system was proposed by Sharma et al. [14] with varying levels of complexity. Although the systems are relatively resilient, they struggle to manage crowded student traffic during the start and end of classes. The system also lacks verification of identity, making it vulnerable to misuse by individuals carrying someone else's RFID card. Rao et al. [15] developed a fingerprint verification system that uses pattern matching technology. This biometric authentication method faces similar challenges as the RFID system and requires more costly equipment. There is also a problem with the hardware system, as different institutions have issues with hardware that they have neglected. If any single endpoint fails, the system becomes unusable. Verma et al. [16] introduced a system that uses facial recognition technology to capture images of students when they enter the classrooms. The captured images are further compared to the existing database to take attendance. They also suggested analyzing data like spending time in the classroom, etc. Asmara et al. [17] also developed a facial recognition system for use in software applications. The authentication system includes identifying human faces that have been authorized access by individuals with permission. An iris recognition-based attendance system has been introduced by Kadry et al. [18] which operates as a wireless network. The paper demonstrates better performance; however, their efficiency varies significantly based on the endpoint where they are developed. Additionally, certain phones lack support for fingerprint scanners, and the speech recognition models vary among different users, leading to potential abuse of the system. Soewito et al. [19] introduced a smartphone-based attendance system that uses fingerprint and speech recognition while also incorporating location and individual attributes for enhanced accuracy. Anil Kumar Sao et al. [20] proposed a template-matching approach for face recognition, addressing issues like pose variations. The method transforms facial images into edge maps and employs template matching using an edge-based technique to represent images in a one-dimensional format. Identification is determined by matching scores. The method employs Local Binary Patterns of Histogram along with the Viola-Jones algorithm to detect and identify student images stored in a dataset. Images that match the criteria are added to the database, while those that cannot be identified are flagged as potential intruders. So, a more efficient and user-friendly attendance system can streamline the process and reduce the risk of inaccuracies.

III. Methodology

In this section, our proposed cloud-based attendance system with leveraging the AWS recognition is presented. The approach for the system has two main parts—student registration and attendance taking. A system is developed by which the registration process can be done and the teacher can easily take attendance by taking only one picture of the whole classroom with a single click. The pictures faces are matched with the existing database and the attendance is taken. The detailed procedure is discussed in this section. The system operates smoothly by integrating several essential AWS services. Amazon S3 is used to securely store all captured images and face data. AWS Rekognition is used to handle facial recognition, matching new faces against the stored face prints for accurate identification. AWS Lambda [21] is utilized to automate tasks like processing images as soon as they are uploaded, ensuring the system runs without manual effort. DynamoDB stores face prints and metadata, providing quick access during recognition. Furthermore, the system sends attendance reports via email, enhancing efficiency and user-friendliness. Finally, three practical images are tested in the system to demonstrate the accuracy of attendance tracking.

A. Registration process

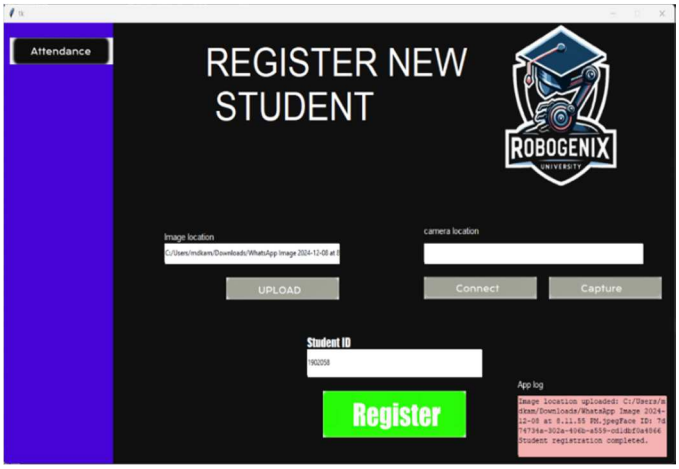


Fig. 1. Designed interface of the registration process,

Authorized licensed use limited to: Chittagon University of Engineering and Technology. Downloaded on July 15,2025 at 21:54:09 UTC from IEEE Xplore. Restrictions apply.

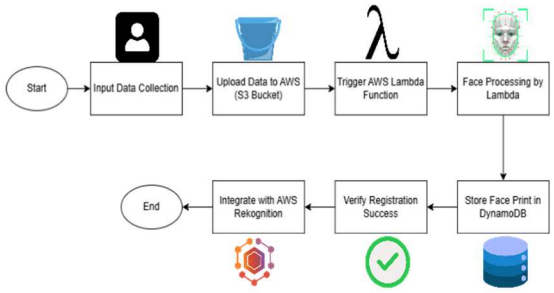


Fig. 2. Flow diagram of the registration process.

Fig. 1 shows the designed interface of the registration system, and Fig. 2 illustrates the overall registration process to add the student identity to the database. After registering any students, the data will be saved to the AWS DynamoDB shown in Fig. 3. The registered student's database will be used to match the face while taking attendance. The overall process involves the following steps:

	A	B
1	RekognitionId	ID
2	7b908854-e00d-405c-ae7c-0e3a971c7e2a	1902002
3	8b07b649-b94b-4323-8291-993f35d2d909	1902003
4	293f7dc7-13a3-4a04-8eab-26f4d6e46ceb	1902006
5	c9e025fd-60bd-45de-9eea-8374592719a9	1902007
6	62971486-1267-4d90-8bce-667040706db0	1902008
7	d3491b92-9335-4672-a1e4-04275ab634ed	1902009
8	bdb378d0-02b2-4152-89a2-0d4a2474eaae	1902010
9	d9005471-7f3d-464e-9383-abbe89400512	1902014
10	5e9ba8d8-879a-4916-a7ff-0d3858cdb7c8	1902015
11	d1314a88-6f2f-45e6-9ad1-af4d30297e6e	1902018
12	f4b63000-ae06-466f-a96a-22ccb0564001	1902019
13	e97d8325-bb29-4121-b300-0c1648c79b0a	1902021
14	83b481a2-f1c4-4fbe-88cc-a34a0e1720bb	1902022
15	fad4c827-324d-4996-b0d7-1465ad6bef0f	1902023
16	5dafc42b-f271-4526-ac26-555a2dfcfc5d	1902025
17	3cdba0e9-94b1-4500-9cd0-93428f0ff76b	1902026
18	4b24b22b-0c08-472c-9234-3723f99a0da8	1902027
19	7a1a9e93-5ccd-4b59-bd05-c97a553321c7	1902030
20	0a89b840-d0fd-41da-a92b-2d4cf88a49a6	1902031
21	d2eea872-9b30-4836-bb4c-d9bf70d0f305	1902034
22	c8ccc65b-c83a-4dff-923e-0d7769b5f497	1902036
23	bcb29dfc-8b01-4fc6-b13a-bd11b1efa708	1902037
24	085b07d4-9249-4edf-ae46-556c1bc51a58	1902038
25	029f0896-a13e-4975-8183-93f28cf63913	1902039
26	b4e171a5-669e-461e-9cf5-c45adaa17015	1902040
27	42073fc8-d2ab-4d4d-a380-ae0216423953	1902041

Fig. 3. Aws DynamoDB Database after all student registered.

(a) Input Data Collection

Initially, clear pictures of the students' faces are taken using the developed app by clicking the capture button from the camera set up in the classroom. A unique identification number is assigned to each student. These inputs are essential to link the student's identity to their facial data for future attendance recognition.

(b) Data Upload to AWS S3

The collected image, along with its unique identification number and metadata such as student ID, name, and class, is uploaded to an Amazon S3 bucket.

(c) Triggering AWS Lambda for Face Processing

Upon uploading the image, an AWS Lambda function is triggered to process it automatically. This function analyzes the image, extracts key facial features, and generates a unique face print, which is a mathematical representation of the student's face. AWS Lambda is a serverless computing service that executes code in response to events, such as image uploads, eliminating the need for manual server management. It automatically adjusts to handle the processing load, making it a reliable and efficient choice for real-time image processing tasks such as facial recognition.

(d) Face Print Storage in DynamoDB

After generating the face print, it and the related metadata are securely stored in Amazon DynamoDB, a fast and scalable NoSQL database. A dedicated table in DynamoDB is created to store and organize face prints along with the related information for each student. Each record in the table contains a unique face print that mathematically represents the student's face, along with the student's ID for easy identification. Additionally, optional information such as the student's name and class can also be stored.

(e) AWS Recognition Integration

The face prints stored in DynamoDB are made accessible to AWS recognition, allowing the system to use the stored data for comparing new faces during the attendance process. When a new image is captured, the recognition system analyzes the face and matches it to the stored face prints. If a match is found, the system identifies the student, confirming their registration and enabling accurate attendance tracking. This streamlined process guarantees that the student is prepared for future attendance checks with minimal manual intervention.

B. Attendance Taking Process

This automated attendance system uses AWS recognition to identify and record the presence of students based on a captured classroom image. The full steps of the system are shown in Fig. 4. The process begins with capturing an image and preprocessing it. A single click takes a full classroom picture, matches the faces, identifies, and separates them. Attendance is then completed by sending an email to the teacher's email ID.

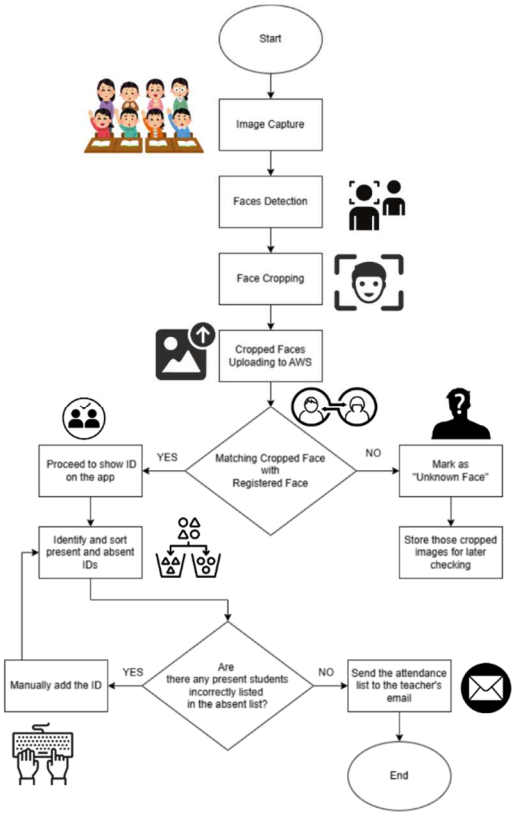


Fig. 4. Proposed method for attendance system.

(1) Image Capture

The teacher needs to capture an image of the entire classroom using a camera positioned to include all students in a single frame.

Authorized licensed use limited to: Chittagon University of Engineering and Technology. Downloaded on July 15,2025 at 21:54:09 UTC from IEEE Xplore. Restrictions apply.

(2) Face Detection

The image is processed to detect faces using convolutional neural networks (CNNs), which are a key component of AWS Recognition. The system detects facial features by analyzing pixel patterns and identifying key landmarks such as the eyes, nose and mouth.

(3) Face Cropping

Detected faces are cropped from the image to create separate images for each individual face. This guarantees precise matching during the recognition phase. The cropped faces are saved temporarily for analysis.

(4) Uploading Cropped Faces to AWS Recognition

The cropped facial images are sent to AWS Recognition via an API call for further analysis. AWS Recognition processes these images in several steps. It first receives cropped face images as input. Then, it extracts detailed facial features and compares them with the face prints stored in DynamoDB. During this feature-matching process, recognition assesses whether the detected face corresponds to any of the registered data. If a match is found, the system retrieves the corresponding student information, including the student ID. If no match is found, the face will be marked as "Unknown" for further review.

(5) Attendance Identification and Sorting

For recognized faces, the system logs the respective student IDs as "Present" and sorts the IDs into "Present" and "Absent" lists. Teachers have the option to manually correct errors, such as
adding IDs for students who were incorrectly marked as absent. Cropped images of unrecognized individuals are stored and shown to the teacher for verification, ensuring the final attendance is accurate.

(7) Sending Attendance Results

Once the recognition and verification processes are successfully completed, the system generates a detailed attendance report. This report includes the IDs of both present and absent students and is sent promptly to the teacher's email for efficient record-keeping. Figure 5 illustrates the email received by the teacher, containing the recorded attendance of the class.

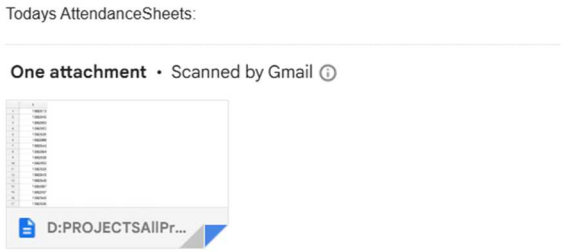


Fig. 5. Mail received by teacher with the full attendance records.

IV. RESULTS AND DISCUSSION

The system has been evaluated for its accuracy in taking attendance by analyzing images of students in various practical scenarios. An industrial tour was organized, during which attendance was recorded and validated using this system. To test its effectiveness, students and teachers were asked to take group photo, which was then analyzed. Teachers were intentionally left unregistered to assess whether the system could distinguish between registered and unknown persons. Additionally, extra student IDs were registered for individuals not present in the photo to verify whether the system could correctly identify absent students.

To evaluate the system, a single image was uploaded as input captured from the camera. The system's user interface is illustrated in Fig. 7. After uploading the image and taking attendance, the list of present students is displayed in log while the list of absent students is shown in log2. If there is a mismatch, the teacher can manually edit attendance after verifying with the class. If unknown faces are detected, their count is shown in the log section and saved in the database for future reference.



Fig. 6. Uploaded image1 for taking the attendance.

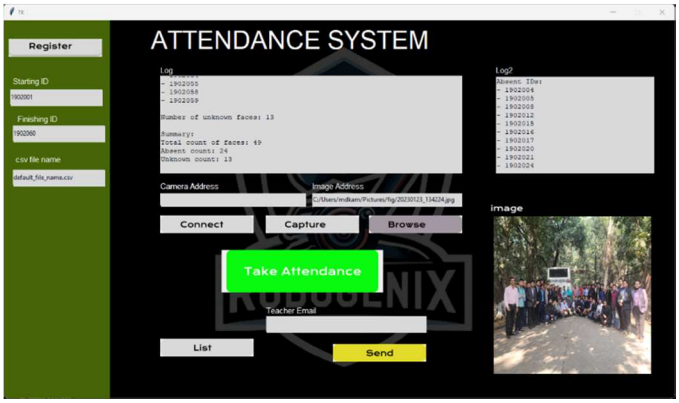


Fig. 7. Detection result of the attendance system for image1.

In this evaluation, an image1 featuring 49 individuals was uploaded, as shown in Fig. 6. Initially, 60 people were registered in the system, with IDs ranging from 190201 to 190260. After analyzing the image, the system detected a total of 49 faces, correctly identifying 36 as registered students marked present and 24 as absent. In image1, 13 individuals were not registered in the system and are identified as unknown, as shown in Fig. 8. These results are 100% accurate as the count perfectly matches with actual value.

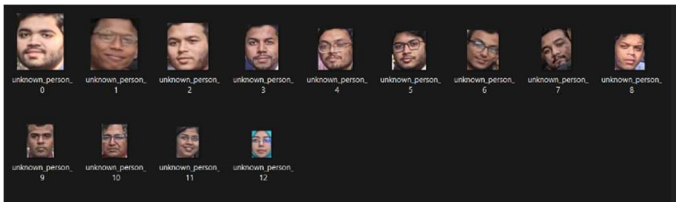


Fig. 8. Stored unknown faces.

Another picture of different scenarios has been uploaded, showing that ideally, 27 students were present while 3 students were absent, whose identities are unknown. The

Authorized licensed use limited to: Chittagon University of Engineering and Technology. Downloaded on July 15,2025 at 21:54:09 UTC from IEEE Xplore. Restrictions apply.

uploaded image2 is shown in Fig. 9, and the outcome of the result is illustrated in the log on the interface in Fig. 10. It successfully detected the 30 faces and accurately matched the number of present people of 27 and absent people of 3 and unknown counted as 3 who were not registered.



Fig. 9. Uploaded image2 for taking the attendance.

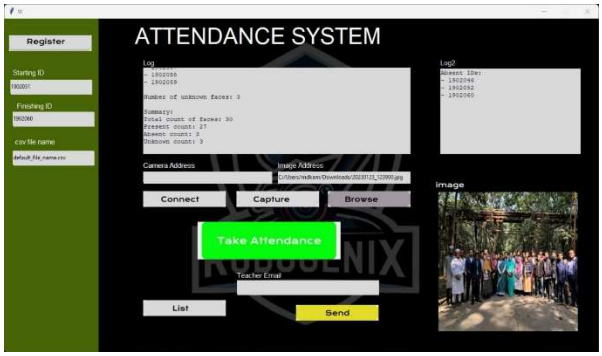


Fig. 10. Detection result of the registered people for image2.

The image3 shown in Fig. 11 is uploaded to the system, which has ideally 14 people presented among the 60 registered people. The system successfully detected the 14 registered people and the other 46 people as absent, as shown in the log of Fig. 12.



Fig. 11. Uploaded image3 for taking the attendance.

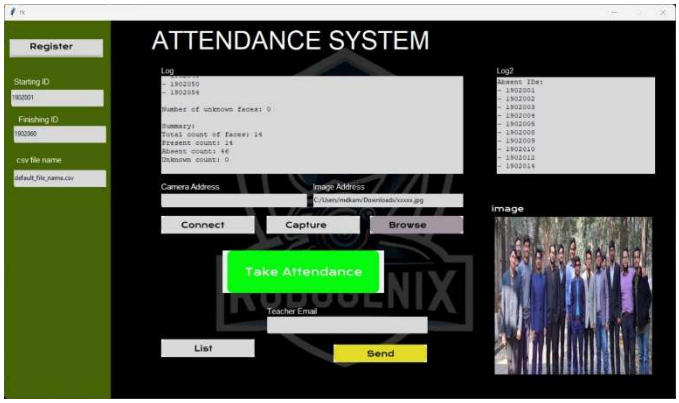


Fig. 12. Detection result of the registered people for image3.

The overall outcomes of the system with a comparison of the actual value are presented in TABLE 01. This table indicates the 100% efficiency of detecting faces of our system and marks the true present, absent, and unknown person.

TABLE 01 : COMPARISON OF ACTUAL AND DETECTED FACE COUNT OF THE ATTENDANCE SYSTEM.

No. of the Fig.	Total face	Detected Face	Actual Present	Detected as present	Actual Absent	Detected as absent
06	49	49	36	36	24	24
09	30	30	27	27	3	3
11	14	14	14	14	46	46

V. CONCLUSION & FUTURE WORK

This study aims to establish an effective cloud-based automated attendance system that utilizes the AWS API for face detection. Amazon S3 is used for securely storing user images, AWS Lambda executes logic in an environment, and DynamoDB stores registered student IDs and face prints while the application sends attendance records via email. The system has the capability to detect faces in real-time with high accuracy. A single captured image containing multiple faces with various expressions can be recognized. It accurately detects all visible faces with 100% precision. This system is designed with criteria to optimize its ability to handle large amounts of data. The proposed system features an excellent graphical user interface, displaying detailed information including the number of present and absent students, as well as the total detected faces, making the system more appealing.

In the future, our system can be improved to operate offline to eliminate the reliance on the Internet and AWS. This will make our system usable in remote areas where the internet is rare. Our system can be improved by implementing continuous video streaming in the app, eliminating the need to capture static images and manually check for missing or incorrectly marked absent students. This system can be further improved to operate fully autonomously by recognizing the teacher and automatically sending lists of present and absent students to the teacher's email.

V. REFERENCES

[1] P. Secreto and E. Tabo, "Impact of Synchronous Class Attendance on the Academic Performance of Undergraduate Students," *International Journal of Educational Management and Development Studies,* vol. 4, no. 1, pp. 109-128, 2023.

[2] M. S. Akbar, P. Sarker, A. T. Mansoor, A. M. Al Ashray, and J. Uddin, "Face recognition and RFID verified attendance system," in *2018 International Conference on Computing, Electronics & Communications Engineering (iCCECE)*, 2018: IEEE, pp. 168-172.

[3] K. O. Okokpujie, E. Noma-Osaghae, O. J. Okesola, S. N. John, and O. Robert, "Design and implementation of a student attendance system using iris biometric recognition," in *2017 International Conference on Computational Science and Computational Intelligence (CSCI)*, 2017: IEEE, pp. 563-567.

[4] S. Tachmammedov, Y. K. Hooi, and K. S. Kalid, "Automated multi-factor analytics for cheat-proofing attendance-taking," in *Proceedings of the 2018 7th international conference on software and computer applications*, 2018, pp. 183-188.

[5] N. Kar, M. K. Debbarma, A. Saha, and D. R. Pal, "Study of implementing automated attendance system using face recognition technique," *International Journal of computer and communication engineering,* vol. 1, no. 2, pp. 100-103, 2012.

[6] J. RoshanTharanga, S. Samarakoon, T. Karunarathne, K. Liyanage, M. Gamage, and D. Perera, "Smart attendance using real time face

Authorized licensed use limited to: Chittagon University of Engineering and Technology. Downloaded on July 15,2025 at 21:54:09 UTC from IEEE Xplore. Restrictions apply.

recognition (smart-fr)," *Department of Electronic and Computer Engineering, Sri Lanka Institute of Information Technology (SLIIT), Malabe, Sri Lanka,* 2013.

[7] K. S. Selvi, P. Chitrakala, and A. A. Jenitha, "Face recognition based attendance marking system," *International Journal of Computer Science and Mobile Computing,* vol. 3, no. 2, pp. 337-342, 2014.

[8] S. Narula and A. Jain, "Cloud computing security: Amazon web service," in *2015 Fifth International Conference on Advanced Computing & Communication Technologies*, 2015: ieee, pp. 501-505.

[9] H. Milhem and N. B. Harrison, "The Quality Attibutes and Architectural Tactics of Amazon Web Services (AWS)," in *2022 Intermountain Engineering, Technology and Computing (IETC)*, 2022: IEEE, pp. 1-6.

[10] V. Sharma, "Object detection and recognition using Amazon Rekognition with Boto3," in *2022 6th International Conference on Trends in Electronics and Informatics (ICOEI)*, 2022: IEEE, pp. 727-732.

[11] R. Ambesh, A. B. Sarfraz, C. Kapoor, H. Joshi, and H. Patel, "Drone Detection using YOLOv4 and Amazon Rekognition," in *2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, 2022: IEEE, pp. 896-902.

[12] N. Kar, M. K. Debbarma, A. Saha, and D. R. Pal, "Study of implementing automated attendance system using face recognition technique," *International Journal of computer and communication engineering,* vol. 1, no. 2, pp. 100-103, 2012.

[13] M. Arsenovic, S. Sladojevic, A. Anderla, and D. Stefanovic, "FaceTime—Deep learning based face recognition attendance system," in *2017 IEEE 15th International symposium on intelligent systems and informatics (SISY)*, 2017: IEEE, pp. 000053-000058.

[14] S. Sharma, S. Shimi, and S. Chatterji, "Radio frequency identification (RFID) based employee monitoring system (EMS)," *International Journal of Current Engineering and Technology,* vol. 4, no. 5, pp. 3441-3444, 2014.

[15] S. Lukas, A. R. Mitra, R. I. Desanti, and D. Krisnadi, "Student attendance system in classroom using face recognition technique," in *2016 International Conference on Information and Communication Technology Convergence (ICTC)*, 2016: IEEE, pp. 1032-1035.

[16] K. G. Oladele, E. J. Iseh, J. O. Oladele, and P. R. Bhaladhare, "AI-Based Face Recognition Attendance System."

[17] R. A. Asmara, M. Ridwan, and G. Budiprasetyo, "Haar cascade and convolutional neural network face detection in client-side for cloud computing face recognition," in *2021 International Conference on Electrical and Information Technology (IEIT)*, 2021: IEEE, pp. 1-5.

[18] S. Kadry and M. Smaili, "Wireless attendance management system based on iris recognition," *Scientific Research and essays,* vol. 5, no. 12, pp. 1428-1435, 2010.

[19] B. Soewito, F. L. Gaol, E. Simanjuntak, and F. E. Gunawan, "Smart mobile attendance system using voice recognition and fingerprint on smartphone," in *2016 International Seminar on Intelligent Technology and Its Applications (ISITIA)*, 2016: IEEE, pp. 175-180.

[20] P. Indhumathi, M. Hirthik, P. Rubesh, and G. P. Kumar, "Face Recognition based secured Attendance System," in *2024 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI)*, 2024: IEEE, pp. 1-5.

[21] S. Gandhi, A. Gore, S. Nimbarte, and J. Abraham, "Implementation and Analysis of a Serverless Shared Drive with AWS Lambda," in *2018 4th International Conference for Convergence in Technology (I2CT)*, 2018: IEEE, pp. 1-6.

Authorized licensed use limited to: Chittagon University of Engineering and Technology. Downloaded on July 15,2025 at 21:54:09 UTC from IEEE Xplore. Restrictions apply.